

Original Image

$$\begin{aligned} & 0 \leq var \left(\frac{1}{nh_n} \sum_{i=1}^n k_1 \left(\frac{Z_{1,i}-s}{h_n} \right) \right) \\ = & \frac{1}{n^2 h_n^2} var \left(\sum_{i=1}^n k_1 \left(\frac{Z_{1,i}-s}{h_n} \right) \right) + \frac{1}{n^2 h_n^2} \sum_{i \neq j=1}^n cov \left(k_1 \left(\frac{Z_{1,i}-s}{h_n} \right), k_1 \left(\frac{Z_{1,j}-s}{h_n} \right) \right) \\ = & \frac{1}{n^2 h_n^2} var \left(\sum_{i=1}^n k_1 \left(\frac{Z_{1,i}-s}{h_n} \right) \right) \quad (\text{since } Z_{1,i} \perp Z_{1,j} \text{ for } i \neq j) \\ = & \frac{1}{nh_n^2} E \left(k_1^2 \left(\frac{Z_1-s}{h_n} \right) \right) - \frac{1}{nh_n^2} \left\{ E \left(k_1^2 \left(\frac{Z_1-s}{h_n} \right) \right) \right\}^2 \\ \leq & \frac{1}{nh_n^2} E \left(k_1^2 \left(\frac{Z_1-s}{h_n} \right) \right) \\ = & \frac{1}{nh_n^2} \int k_1^2 \left(\frac{Z_1-s}{h_n} \right) f_{Z_1}(z_1) dz_1 = \frac{1}{nh_n} \int k_1^2(m) f_{Z_1}(s + mh_n) dm \\ = & \frac{1}{nh_n} \int k_1^2(m) \left[f_{Z_1}(s) + mh_n f'_{Z_1}(\xi_n) \right] dm \quad (\text{here } \xi_n \in (s, s + mh_n)) \\ = & \frac{f_{Z_1}(s)}{nh_n} \int k_1^2(m) dm + \frac{f'_{Z_1}(\xi_n)}{n} \int mk_1^2(m) dm \end{aligned}$$

Visualized Result

$$\begin{aligned} & 0 \leq var \left(\frac{1}{nh_n} \sum_{i=1}^n k_1 \left(\frac{Z_{1,i}-s}{h_n} \right) \right) \\ = & \frac{1}{n^2 h_n^2} var \left(\sum_{i=1}^n k_1 \left(\frac{Z_{1,i}-s}{h_n} \right) \right) + \frac{1}{n^2 h_n^2} \sum_{i \neq j=1}^n cov \left(k_1 \left(\frac{Z_{1,i}-s}{h_n} \right), k_1 \left(\frac{Z_{1,j}-s}{h_n} \right) \right) \\ = & \frac{1}{n^2 h_n^2} var \left(\sum_{i=1}^n k_1 \left(\frac{Z_{1,i}-s}{h_n} \right) \right) \quad (\text{since } Z_{1,i} \perp Z_{1,j} \text{ for } i \neq j) \\ = & \frac{1}{nh_n^2} E \left(k_1^2 \left(\frac{Z_1-s}{h_n} \right) \right) - \frac{1}{nh_n^2} \left\{ E \left(k_1^2 \left(\frac{Z_1-s}{h_n} \right) \right) \right\}^2 \\ \leq & \frac{1}{nh_n^2} E \left(k_1^2 \left(\frac{Z_1-s}{h_n} \right) \right) \\ = & \frac{1}{nh_n^2} \int k_1^2 \left(\frac{Z_1-s}{h_n} \right) f_{Z_1}(z_1) dz_1 = \frac{1}{nh_n} \int k_1^2(m) f_{Z_1}(s + mh_n) dm \\ = & \frac{1}{nh_n} \int k_1^2(m) \left[f_{Z_1}(s) + mh_n f'_{Z_1}(\xi_n) \right] dm \quad (\text{here } \xi_n \in (s, s + mh_n)) \\ = & \frac{f_{Z_1}(s)}{nh_n} \int k_1^2(m) dm + \frac{f'_{Z_1}(\xi_n)}{n} \int mk_1^2(m) dm \end{aligned}$$

Parsed Result

$$\begin{aligned} & 0 \leq var \left(\frac{1}{nh_n} \sum_{i=1}^n k_1 \left(\frac{Z_{1,i}-s}{h_n} \right) \right) \\ &= \frac{1}{n^2 h_n^2} var \left(\sum_{i=1}^n k_1 \left(\frac{Z_{1,i}-s}{h_n} \right) \right) + \frac{1}{n^2 h_n^2} \sum_{i \neq j=1}^n cov \left(k_1 \left(\frac{Z_{1,i}-s}{h_n} \right), k_1 \left(\frac{Z_{1,j}-s}{h_n} \right) \right) \\ &= \frac{1}{n^2 h_n^2} var \left(\sum_{i=1}^n k_1 \left(\frac{Z_{1,i}-s}{h_n} \right) \right) \quad (\text{since } Z_{1,i} \perp Z_{1,j} \text{ for } i \neq j) \\ &= \frac{1}{nh_n^2} E \left(k_1^2 \left(\frac{Z_1-s}{h_n} \right) \right) - \frac{1}{nh_n^2} \left\{ E \left(k_1^2 \left(\frac{Z_1-s}{h_n} \right) \right) \right\}^2 \\ &\leq \frac{1}{nh_n^2} E \left(k_1^2 \left(\frac{Z_1-s}{h_n} \right) \right) \\ &= \frac{1}{nh_n^2} \int k_1^2 \left(\frac{Z_1-s}{h_n} \right) f_{Z_1}(z_1) dz_1 = \frac{1}{nh_n} \int k_1^2(m) f_{Z_1}(s + mh_n) dm \\ &= \frac{1}{nh_n} \int k_1^2(m) \left[f_{Z_1}(s) + mh_n f'_{Z_1}(\xi_n) \right] dm \quad (\text{here } \xi_n \in (s, s + mh_n)) \\ &= \frac{f_{Z_1}(s)}{nh_n} \int k_1^2(m) dm + \frac{f'_{Z_1}(\xi_n)}{n} \int mk_1^2(m) dm \end{aligned}$$